

Project 3: Autonomous Medical Billing Code Team (A2A)

This project focuses on simulating a team of specialist agents that converse with each other to solve a complex information extraction and coding task.

- **Objective:** To build an autonomous multi-agent system that can read unstructured clinical notes, extract relevant medical information, assign standardized billing codes (ICD-10), and draft a billing report.
- **Core Framework: AutoGen (A2A)**
- **Real-World Scenario:** A hospital's billing department spends hours manually reading doctor's notes to assign the correct billing codes. This project aims to automate the initial draft of this process, flagging ambiguities for human review.
- **Key Tasks & Steps:**
 1. **Agent Setup:**
 - Configure a UserProxyAgent to represent the human user who initiates the task and provides the clinical note.
 - Create specialized AssistantAgents:
 - **Clinical_Extractor:** Prompted to read medical text and identify all diagnoses, symptoms, and procedures.
 - **ICD_10_Coder:** Given a list of medical terms, its job is to find the corresponding ICD-10 codes. It can use a tool (web search via Serper or searching a provided CSV/PDF of codes).
 - **Report_Generator:** Takes the structured list of terms and codes to format a final billing summary.
 2. **Collaborative Conversation:**
 - Initiate a chat where the UserProxyAgent provides the clinical note to the Clinical_Extractor.
 - The Clinical_Extractor then presents its findings to the ICD_10_Coder.
 - The conversation continues until a final, structured list of codes is produced and passed to the Report_Generator.
- **Tech Stack:**
 - **Agent Framework:** Microsoft Autogen
 - **Tools:** Python, Serper API, Pandas (for handling coding sheets).
 - **LLM:** An OpenAI model is recommended, as Autogen is highly optimized for it.
 - **Evaluation:** Opik or Langfuse to monitor the agent-to-agent conversations. Evaluate the **accuracy** of the assigned codes against a "ground truth" set of manually coded notes.